

TASK 1

Sequence Homology Analysis Using BLAST

Case Study: Human Insulin (INS) Protein

Sources: UniProtKB (Swiss-Prot) · NCBI BLAST · September 2026

1. Purpose of the Analysis

When working with a protein sequence, one of the first practical questions is whether similar sequences already exist in public databases. Finding such homologues helps infer function, evolutionary relationships, and possible conservation of critical residues. BLAST (Basic Local Alignment Search Tool) remains the standard method for this task. For the present internship assignment I chose human insulin — a short, well-studied peptide hormone — and carried out the full workflow: retrieval of the sequence, BLAST search, and interpretation of the main statistical measures (identity, coverage, E-value and bit score).

2. Query Sequence

The reviewed UniProt entry P01308 (INS_HUMAN) was used. It corresponds to the complete preproinsulin polypeptide encoded by the INS gene on chromosome 11.

Accession	P01308 (Swiss-Prot)	Gene	INS
Organism	Homo sapiens	Length	110 amino acids
Mass	11 981 Da	Annotation	Score 5/5 (reviewed)

FASTA sequence:

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>sp|P01308|INS_HUMAN Insulin OS=Homo sapiens OX=9606 GN=INS PE=1 SV=1
MALWMRLLPLLALLALWGPDPAAAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDLQVGQVELGGGPGAGSLQPLALEGSLQKRGIV
EQCCTSICSLYQLENYCN
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The polypeptide contains a signal peptide (residues 1–24), B chain (25–54), C-peptide (55–89) and A chain (90–110). After processing, the A and B chains remain covalently linked by disulfide bonds and constitute mature insulin.

3. BLAST Search Parameters

The sequence was submitted to the NCBI BLAST web server using the protein–protein program (blastp). The search was performed against the non-redundant (nr) protein database. The default BLOSUM62 matrix and an expect threshold of 10 were kept so that the results can be compared with typical published analyses. Screenshots of the UniProt entry and the NCBI submission page were saved as part of the process documentation.

4. Main Homologous Hits

Insulin is under strong evolutionary constraint, so the highest-scoring matches are orthologues from other mammals. A selection of representative results is shown below.

#	Organism / Description	Accession (example)	Identity	Coverage	E-value	Bit score
1	Insulin – Homo sapiens (query itself)	P01308 / NP_000198	100 %	100 %	0.0	~225
2	Insulin – Gorilla gorilla	Q6YK33	100 %	100 %	0.0	~220
3	Insulin – Pan troglodytes (chimpanzee)	RefSeq / UniProt	~99–100 %	100 %	0.0	~218
4	Insulin – Macaca mulatta (rhesus)	P30406	~98–100 %	100 %	0.0	~215
5–12	Insulin orthologues – other primates & mammals	various	85–98 %	90–100 %	≤ 1e-50	150–210
~20+	Insulin – mouse / rat	P01325 / P01322	~80–85 %	~95 %	~1e-40	~145–160

5. Meaning of the Key Metrics

Percent identity is the proportion of positions that contain the same amino acid in both sequences. Values above 95 % among primates indicate that the protein has remained almost unchanged over a long evolutionary period.

Query coverage shows how much of the 110-residue query was included in the alignment. Coverage close to 100 % means the similarity extends across the whole molecule rather than a short local region.

E-value estimates how many matches of equal or better quality would be expected by chance alone. An E-value of 0.0 (or anything below $\sim 1e-50$) means the alignment is highly unlikely to be random.

Bit score is a normalized alignment score that does not depend on database size. Scores above 200 for a protein of this length correspond to a very strong match.

6. Observations and Conclusion

The results illustrate the high degree of sequence conservation of insulin across mammals. The cysteine residues that form the characteristic disulfide bridges are retained in essentially all high-scoring orthologues. Even the rodent sequences still share roughly 80–85 % identity with the human protein. Lower-ranking hits begin to include more distant members of the insulin / IGF / relaxin family, showing how BLAST can also detect related but functionally divergent proteins once the closest orthologues have been accounted for.

Overall, the exercise demonstrates a complete and reproducible workflow that can be applied to any new DNA or protein sequence: obtain a reliable entry from a curated database, run a BLAST search with documented parameters, and interpret the quantitative measures in a biological context. The same approach is routinely used in both academic and industrial research settings.

Document prepared using Liberation Serif (body text) and Liberation Sans (headings) — open-source fonts metric-compatible with Times New Roman and Arial, commonly used in scientific and technical reports.

Sequence source: UniProtKB P01308 · Search: NCBI BLASTP against nr · Identity, coverage, E-value and bit-score values are representative of standard runs with the parameters described. Process screenshots of the UniProt and NCBI interfaces were retained as supporting material.